

NAME

mq_send, mq_timedsend – send a message to a message queue

LIBRARY

Real-time library (*librt*, *-lrt*)

SYNOPSIS

```
#include <mqueue.h>
```

```
int mq_send(mqd_t mqdes, const char msg_ptr[msg_len],
            size_t msg_len, unsigned int msg_prio);
```

```
#include <time.h>
```

```
#include <mqueue.h>
```

```
int mq_timedsend(mqd_t mqdes, const char msg_ptr[msg_len],
                 size_t msg_len, unsigned int msg_prio,
                 const struct timespec *abs_timeout);
```

Feature Test Macro Requirements for glibc (see **feature_test_macros(7)**):

```
mq_timedsend():
```

```
_POSIX_C_SOURCE >= 200112L
```

DESCRIPTION

mq_send() adds the message pointed to by *msg_ptr* to the message queue referred to by the message queue descriptor *mqdes*. The *msg_len* argument specifies the length of the message pointed to by *msg_ptr*; this length must be less than or equal to the queue's *mq_msgsize* attribute. Zero-length messages are allowed.

The *msg_prio* argument is a nonnegative integer that specifies the priority of this message. Messages are placed on the queue in decreasing order of priority, with newer messages of the same priority being placed after older messages with the same priority. See **mq_overview(7)** for details on the range for the message priority.

If the message queue is already full (i.e., the number of messages on the queue equals the queue's *mq_maxmsg* attribute), then, by default, **mq_send()** blocks until sufficient space becomes available to allow the message to be queued, or until the call is interrupted by a signal handler. If the **O_NONBLOCK** flag is enabled for the message queue description, then the call instead fails immediately with the error **EAGAIN**.

mq_timedsend() behaves just like **mq_send()**, except that if the queue is full and the **O_NONBLOCK** flag is not enabled for the message queue description, then *abs_timeout* points to a structure which specifies how long the call will block. This value is an absolute timeout in seconds and nanoseconds since the Epoch, 1970-01-01 00:00:00 +0000 (UTC), specified in a **timespec(3)** structure.

If the message queue is full, and the timeout has already expired by the time of the call, **mq_timedsend()** returns immediately.

RETURN VALUE

On success, **mq_send()** and **mq_timedsend()** return zero; on error, *-1* is returned, with *errno* set to indicate the error.

ERRORS**EAGAIN**

The queue was full, and the **O_NONBLOCK** flag was set for the message queue description referred to by *mqdes*.

EBADF

The descriptor specified in *mqdes* was invalid or not opened for writing.

EINTR

The call was interrupted by a signal handler; see **signal(7)**.

EINVAL

The call would have blocked, and *abs_timeout* was invalid, either because *tv_sec* was less than zero, or because *tv_nsec* was less than zero or greater than 1000 million.

EMSGSIZE

msg_len was greater than the *mq_msgsize* attribute of the message queue.

ETIMEDOUT

The call timed out before a message could be transferred.

ATTRIBUTES

For an explanation of the terms used in this section, see **attributes(7)**.

Interface	Attribute	Value
mq_send() , mq_timedsend()	Thread safety	MT-Safe

VERSIONS

On Linux, **mq_timedsend()** is a system call, and **mq_send()** is a library function layered on top of that system call.

STANDARDS

POSIX.1-2008.

HISTORY

POSIX.1-2001.

SEE ALSO

mq_close(3), **mq_getattr(3)**, **mq_notify(3)**, **mq_open(3)**, **mq_receive(3)**, **mq_unlink(3)**, **timespec(3)**, **mq_overview(7)**, **time(7)**